1.4.3 Variable Versions and Suffixes

The V-Dem Dataset contains several versions of the variables coded by country experts (type C variables).

- **Model Estimates**
  "Model Estimates" — Measurement Model Output:
  This version has no special suffix (*e.g.* v2elmulpar). This version of the variables provides country-year (country-date in the alternative dataset) point estimates from the V-Dem measurement model (see Pemstein et al. (2026)). The measurement model aggregates the ratings provided by multiple country experts and, taking disagreement and measurement error into account, produces a probability distribution over country-year scores on a standardized interval scale (see the *V-Dem Methodology* document). The point estimates are the median values of these distributions for each country-year. The scale of a measurement model variable is similar to a normal ("Z") score (*e.g.* typically between -5 and 5, with 0 approximately representing the mean for all country-years in the sample) though it does not necessarily follow a normal distribution. To aid interpretation, the codebook entries for "C" variables report the approximate ranges of the continuous scores that correspond to each of the survey scale scores. For most purposes, these are the preferred versions of the variables for time series regression and other estimation strategies.

  "Model Estimates Measure of Uncertainty" — Measurement Model Highest Posterior Density (HPD) Intervals:
  This version has the suffixes: "codelow" and "codehigh" (*e.g.* v2elmulpar_codelow and v2elmulpar_codehigh). These two kinds of variables ["code low" and "code high"] demarcate the interval in which the measurement model places 68 percent of the probability mass for each country-year score, which is approximately equivalent to one standard deviation upper and lower bounds. If the underlying posterior distribution is skewed, the HPDs reflect this with unequal distances between the point estimate and the high and low estimates. We also provide a standard calculation for standard deviation which is marked with the suffix "sd" (*e.g.*, v2elmulpar_sd). The SD might be used to compute the standard frequentist confidence intervals. For further explanation, see the "Comparison of Model Estimates, Original Scale, and Ordinal Scale" section below.

- **Original Scale (*_osp)**
  "Original Scale" — Linearized Original Scale Posterior Prediction:
  This version has the suffix "_osp," (*e.g.* v2elmulpar_osp). In this version of the variables, we have linearly translated the measurement model point estimates back to the original ordinal scale of each variable (*e.g.* 0-4 for v2elmulpar_osp) as an interval measure. The decimals in the _osp version roughly indicate the distance between the point estimate from the linearized measurement model posterior prediction and the threshold for reaching the next level on the original ordinal scale. Thus, a _osp value of 1.25 indicates that the median measurement model posterior predicted value was closer to the ordinal value of 1 than 2 on the original scale. Technically, it calculates the sum of the posterior probabilities that the estimate is in a particular category: If a particular country-year-variable has a probability of 90% to be in category "4", a 10% probability of being in category "3", and 0% probability of being in categories "2", "1", and "0", the result is a value of 3.9 ($4 * 0.9 + 3 * 0.1 = 3.6 + 0.3$). Since there is no conventional theoretical justification for linearly mapping ordinal posterior predictions onto an interval scale, these scores should primarily be used for heuristic purposes. Using the "Ordinal Scale" estimates—or incorporating the properties of ordinal probit models into the estimation procedure—is thus preferable to using the _osp estimates in statistical analyses. However, since the _osp version maps onto the coding criteria found in the V-Dem Codebook, and is strongly correlated with the Measurement Model output (typically at .98 or higher), some users may find the _osp version useful in estimating quantities such as marginal effects with a clear substantive interpretation. If a user uses _osp data in statistical analyses it is imperative that they confirm that the results are compatible with estimations using Measurement Model output.

  "Original Scale Measure of Uncertainty" — Linearized Original Scale HPD Intervals:
  This version has the suffixes "codelow" and "codehigh" (*e.g.* v2elmulpar_osp_codelow and